

Data Practices of Cross-Domain Integration: Draw-and-Write Interviews with Interdisciplinary Scientists

Abstract

Cross-domain integration is generally poorly documented and evaluated indirectly—we know that interdisciplinary scientists work with data from multiple domain contexts, which differ in form, structure, and terminology, but not about how these data are synthesized. In this paper, I explore the practice of cross-domain data integration by completing ethnographic draw-and-write interviews about data practices at interdisciplinary science laboratories. I observe that cross-domain integration is conceived of by scientists as a data practice, but that the activities comprising this process are not formalized in the way that other research practices, like collection or domain-standard statistical analyses, are. Scientists struggle to explain how their laboratories integrate data from different domains; this uncertainty applies to integration and not to data practices more generally. I suggest that interdisciplinary scientists conceive of cross-domain integration as trivial, data centric ‘grunt work’, in service to and distinct from ‘real’ scientific research, and discuss adverse impacts of this perspective on reproducibility and on rigorous evaluation of interdisciplinary data science.

Keywords

Data practices, Data cultures, Ethnography, Laboratory studies, Integration, Interdisciplinary Science

Background

Broad Data

When data are big, they are also broad. A project using big data is often a project using data that are generated within multiple domain contexts—that is, ‘broad’ data—which take on different forms, are evaluated by different quality criteria, and adhere to different standards of documentation according to the norms of those domains (Shah et al., 2023; Hendler, 2014). Domains can include academic disciplines, communities of practice, and professional sectors;

groups with shared ontological and epistemological commitments (Fry and Talja, 2007; Hjørland, 2017). In recent years, data and computer science scholars have adopted domain analytic approaches to explain data practice. For example, Ribes (2019) characterizes domains according to data formats, models, and schemas, as well as ‘fuzzier’ semantic features of data such as domain-specific metadata, terminology, and implicit assumptions, and others have explored how data practice differs in field-specific contexts (for example, see Hassett 2018; Donatz-Fest, 2024).

The touted benefits of big and broad data include the ability for scientists to make use of data from multiple sources and thus conduct more holistic analyses, enjoy increased coverage, and see past domain-specific ‘blind spots’ (Housley et al., 2014; Feters and Molina-Azorin, 2017). In practice, it is not so straightforward to take advantage of broad data. It is not sufficient to merely gather large quantities of data from several domains, because each of these domains has shaped the data to meet certain criteria and take certain forms in accordance with that domain’s conventions. Attempts to integrate (alternately: synthesize, combine, converge) these broad data across domain boundaries result in friction due to differences in terminology, scale, error margins, and preferred methods of analysis (Edwards et al., 2011; Panagiotidou et al. 2022). To put it another way: cross-domain integration is a non-trivial undertaking, something that takes effort and impacts research outputs.

Cross-Domain Data Integration

We have seen that cross-domain integration is something important that happens during research involving broad data. Our understanding of what cross-domain integration actually *is*, that is, of the specific activity or activities being undertaken, is less certain, because it is conceived of entirely differently in different areas of study. We can observe two broad ‘shapes’ of cross-domain integration described in the literature: technical integration and social integration.

Computer science, data science, and statistics scholars tend to describe cross-domain integration in technical, object-focused terms. Such approaches have described integration as a process of ‘quantizing’ qualitative data generated in one domain setting so it can be incorporated into statistical models in others (Anguera et al., 2022; Batley, 2022; Shi et al., 2023), as implementing interdisciplinary databases capable of storing and linking data according to multiple domain standards (Pelizza, 2021), or as developing algorithms (Xu and Yu, 2017)

and benchmarking protocols (Bashardoost et al., 2019) to structure exchange between heterogeneous datasets and knowledge bases. The defining characteristic of these technical conceptions of cross-domain integration is a focus on data artefacts and data-as-material, with little inquiry into what happens once the data are stored together or transformed into compatible formats—it is assumed implicitly that technical integration leads to effective use and conceptual understanding of broad data from across domains.

In a parallel and mostly non-overlapping discussion, scholars of pedagogy, science and technology studies, and information behaviour instead frame cross-domain integration in terms of collaboration, of data sharing on a more social, human level. These scholars propose theoretical frameworks for how people integrate (Schoonenboom, 2022; Boon and Van Baalen, 2019), examine practices that help and hinder collaboration for teams of scientists from multiple domains (Palmer, 2013; Parti and Szigeti, 2021), and conduct bibliometric studies of published research to demonstrate how cross-domain scholars cite and collaborate outside of their ‘home domains’ (Hu and Zhang, 2017; Laursen et al., 2022). These social framings of cross-domain integration tend to be either exclusively theoretical (not empirically validated) or retrospective (empirically studied *after* research has been completed, relying on people’s recollection or on published Methods sections of research articles).

Though broadly about the same phenomenon (that is: the synthesis of data from multiple domains), technical and social framings of cross-domain integration are largely insulated from each other. This is our first need for an alternate approach: critical data scholars have demonstrated that data are sociomaterially constructed, products of the interaction of human actors *with* material data artefacts (Thomer and Wickett, 2020; Bates et al., 2016). Put succinctly, “the social and material are inseparable, neither taking precedence over the other” (Donatz-Fest, 2024), and to understand the activities of cross-domain integration, we must consider both. Existing approaches to studying integration also share weaknesses: in looking at technical or social forms of integration, studies tend to implicitly assume that one leads to the other without inquiring about why or how. This, along with the way that most empirical studies of integration are retrospective and/or based on citation networks of final publications, means that the actual activities of cross-domain integration undertaken during research are rarely articulated or documented, and thus become ‘invisible’ or ‘forgotten’ (Boix Mansilla, 2006; Wagner et al., 2011; Laursen et al., 2022). This is problematic: **how can cross-domain, broad data research be trusted if a major part of the work (integration) is not documented and made subject to evaluation and review?**

Open Questions

To address these gaps in the present understanding of how broad data are integrated, I suggest framing cross-domain integration as a data practice. By problematizing integration as a practice (rather than behaviour or more granular activities), we position the data work that comprises integration as something situated in the social context of a domain or domains (Savolainen, 2007), something learned, potentially routine, and subject to rigorous critique (Reckwitz, 2002; Schatzki et al., 2001). We can find precedent for this approach in emerging scholarship justifying the theory and empirically demonstrating its application (Blok et al., 2017; Christin, 2017; Thomer, 2022). In adopting this approach, I take methodological inspiration from Bates et al.'s (2016) data journeys approach, and from Star and Griesemer's (1989) institutional ethnography, to center data-as-document, and as boundary object. That is, I suggest that by 'following the data' and how scientists engage with it during cross-domain research, we will be able to observe how data transverse and transcend domain boundaries, and to understand practices of cross-domain integration that may not be captured in formal methods sections or research protocols, thereby addressing gaps in understanding from existing work on the topic.

Using this framing, I completed ethnographic draw-and-write interviews at interdisciplinary research laboratories to explore the questions:

- A) How do scientists' data practices shape their goals of cross-domain integration? *And, vice versa,*
- B) How do cross-domain research goals shape scientists' data practices?

Methods and Data

Data Collection

The laboratory has long been a study site for exploring the sociomaterial practices of science (Latour and Woolgar, 1979). Participants in this study are members of data-intensive interdisciplinary laboratories within existing research institutions. Here we define 'data intensive science laboratories' as laboratories where research involves a significant data science and/or computational component in addition to traditional 'wet lab' research; participants self-identified as interdisciplinary researchers. 'Existing research institutions' include universities and hospital- or government- affiliated laboratories. Participants were recruited purposively (given the

relatively limited amount of individuals who meet the inclusion criteria) from laboratories within the Greater Toronto Area, to facilitate in-person visits to laboratory locations. In total, five participants were interviewed. This sample size is not sufficiently large to allow for generalizable quantitative results, but is conventional and optimal for the qualitative, 'thick' description of each laboratory being sought here, and the large volume of data generated from each interview.

Data were collected through one-on-one, semi-structured 'draw and write' ethnographic interviews with participants, following Hartel (2019). Interviews took place at each participant's laboratory site and were audio recorded, with the interviewer taking field notes to record context not captured in recordings. Questions were intentionally open-ended, proceeding roughly through the following thematic sections pertaining to the research topic:

1. People (Questions about domain backgrounds of scientists in the laboratory, challenges in bringing together different roles)
2. Data workflow (Questions aimed at understanding what kinds of data the laboratory generates, what technology are used, and how different domain data are integrated)
3. Context (Questions about participants' views on broader trends in the field)

During section two, participants were invited to sketch and narrate a drawing of their laboratory's data workflow (adopting best practices from Brailas' 2020 paper and Hartel (2019)). The physical drawing was retained for analysis, and audio transcriptions transcribed. In transcripts and discussion of findings in this paper, participants and laboratory sites are referred to by generic pseudonyms to support anonymity (e.g. 'Participant A', 'Site 1').

Data Analysis

Interview transcripts and field notes are analyzed using thematic analysis as described by Braun and Clarke (2012). This process is neither wholly deductive nor wholly inductive; rather, in the joint approach demonstrated by Byrne (2022), I explore text data with sensitizing concepts (i.e. existing theories of integration) in mind, but with openness to finding the as-yet-unknown themes that occur 'around' the subject. Analysis of workflow drawings has fewer consensus methodological precedents, and instead draws on a range of established approaches: I take inspiration from 'conceptual modeling' of socio-technical workflows to gain insight about process and practices (Nalchigar and Yu, 2018; Recker et al., 2021), from Decuyper's (2020) 'visual network analysis' (particularly ideas of 'interfaces' between actors or

data objects in a workflow as sites where cross-domain integration may be occurring), and from the visual analysis methods described by Hartel (2017) for analyzing hand-drawn diagrams and graphically organized written text.

Findings

“It’s A Lot”: Struggles of Cross-Domain Work

In existing scholarship on interdisciplinarity and cross-domain collaboration, certain themes such as funding challenges, issues with communication, and a sense of isolation from traditional academic communities, occur repeatedly (Manathunga et al., 2009; Shah et al., 2023). The present study is no exception. Four out of five participants (all but Participant C) raised the subject of funding without prompting, and three explicitly linked difficulties in being awarded funding to the cross-domain nature of their research. Per Participant E, the work of interdisciplinarity is “actually very valuable, but I don’t know how valued it is”, while Participant B expressed that “incentives are often not aligned for multidisciplinary collaborations.” Participant A diagnosed the issue further by saying that, “lots of funding bodies say they want new big ideas, but they don’t. They want incremental science. And I guess what I would say is I feel like I’ve built a career in thumbing my nose at people who want incremental science.” These challenges do not stop at obtaining funding: scientists expressed challenges with communication, particularly with ‘translation’ of domain-specific terminology, and a sense of isolation from their academic communities. Participant E said that their cross-domain research means they have “imposter syndrome in two disciplines” due to being an expert at neither, and Participant A repeated, “as I say to people over and over again, I have no idea what I’m doing.”

I begin by reporting these findings not because they are groundbreaking—as stated, they support similar findings from elsewhere in the literature—but because, though not directly data-related, these concerns from participants serve as important context for our exploration of cross-domain data work. Our interviews show that the scientists conducting cross-domain research are doing so with a sense of ‘breaking new ground’, doing something without an existing standard or correct approach. This lack of procedural certainty and precedent manifests in how participants understand (or *don’t* understand) the data practices of integration.

The Integration Gap

Participants were asked to sketch out their laboratories' data workflows with the goal of identifying the part(s) of the process where cross-domain integration happens. While completing these drawings, participants provided extremely detailed descriptions—verbally and in their workflow sketches—of how and from where they produced or collected their data, as well as the formats data took in its raw form and after pre-processing (generally text files, spreadsheets, and images). Most participants were able to describe the software used for data analysis, usually Python, R, Julia, or proprietary, domain-specific tools, and the ways that their lab stores data (e.g. institutionally or commercially-hosted cloud servers, SQL databases). Some offered to show examples of specific spreadsheets or database front-ends and walked the interviewer through their data materials in even more detail. Clearly, scientists at cross-domain laboratories, even when they took care in our interviews to emphasize that they are not data scientists or computer scientists, have strong data literacy and possess vocabularies with which to discuss their data practices. This makes it all the more noteworthy that **all participants struggled to articulate how they integrate data from across domains.**

[insert Figure 1]

Fig. 1. Workflow sketches from Participants D, C, and B (clockwise from top left). Red arrows point at 'integration' components of data workflows. Redacted components of images (covered by black rectangles) contain names of individuals or organizations.

When examining data workflow sketches, we can follow from Decuyper's (2020) strategies for visual network analysis and focus our attention on 'interfaces', or sites where distinct actors and data objects in a workflow come together. These interfaces are of interest to us as steps in the research process where data become boundary objects and cross-domain integration is likely to be occurring. In the data workflows drawn by participants, data collection steps tended to be described in domain-specific terms (e.g. "sociological data") until a single integration interface (as opposed to integration as an ongoing or multi-step practice) that usually occurred toward the end of research data workflows. Where prior data collection and curation steps included additional detail in words (e.g. file types, explanations of data sources) or illustrative drawings, integration steps often stood alone. As shown in Figure 1, Participant C

wrote 'INTERPRETATION', Participant B drew a human stick figure and described it aloud as "doing the analysis" and "developing new tools", and Participant D wrote and circled the word 'STORY'. When asked to elaborate on these integration stages of their data workflows, participants lacked the certainty with which they had discussed the rest of their lab's data work. Taking Participant D as a particularly colourful example, they initially narrated the 'STORY' step in their workflow sketch as a "compression step, I guess, where we integrate these and we think about story" (evocative of ideas of cross-domain integration as compression described and disputed elsewhere; see Anguera et al. (2022) and Rovane (2010)). When asked to elaborate on what they meant by 'compressing' and 'integrating' data, the scientist replied,

"I don't know, haven't done it yet on this project... um... I'm terrified of this. I really don't know how to do it and there's more than one student and so you think about like, who takes ownership, and do you like have each student lead one substory? Uh, so there's problems- there's tensions of methods and ownership, um, and different disciplinary norms about paper length, authorship styles, authorship lists, and... uh... [long pause] I'm trying to think of an example I have for you, where I have actually published a mixed methods something... [laughs] I don't have any."

Participant D then walked the interviewer through a second data workflow of another project from their lab, and again, when prompted to elaborate about the integration interface, struggled to illustrate or describe how different domain data components would be brought together, concluding that,

"certainly I think my data handling methods would be well characterized by 'haphazard' and certainly not explicit or considered or purposeful or anything like that, and that would not be offensive in the slightest."

This lack of defined practice for integrating data and data work from different domains was characteristic across all interviews, and reflective of concerns from critics of cross-domain research that, even in the most rigorous research settings, the absence of a documented, formalized approach to integration means that *this component* of cross-domain research workflows cannot be properly evaluated and thus lacks rigour (Graff, 2016; Jacobs and Frickel, 2009; Pasipamire and Masuku, 2022). The question then becomes: if cross-domain integration is not at present a formal research practice, what *informal* activities are taking place during this stage of research?

One approach (taken by Participant D) is to have different lab members each work on ‘substories’ or individual components of a research project most relevant to their domain training then combine completed findings in a shared final publication. A potential issue with this approach is that, if research data are being collected, analyzed, and interpreted in domain-specific ‘substories’, with any integration only occurring during the writing of a final paper, it is unclear whether impactful cross-domain exchange is even taking place. Indeed, this concern came up during the interviews, when Participant D spoke about how despite their intentions, data from one domain did not significantly impact or interact with data from the other. The second, most common approach to cross-domain integration offered by participants was to avoid it, or at least to minimize it, while still framing their research as interdisciplinary. Participants A and B described how projects are categorized into one of the domains relevant to their labs, and the team member with expertise in that domain takes the lead and situates the (still ostensibly interdisciplinary) project in the chosen domain. Explanation from Participant E may provide insight as to why this strategy is adopted: it goes, as much interdisciplinary research does, back to the issue of funding. When applying for grants, said Participant E, “I have to always kind of bend it a little bit and be like, this is the physics part of my research, or this is the stats part of my research, and like, write it in that way, because there isn't something that fits just right.” This phenomenon, where one domain dominates what is intended as a cross-domain endeavour, is called ‘disciplinary capture’, and can limit the benefits cross-domain research is supposed to provide (Brister, 2016).

To summarize, we can take away that cross-domain integration is something that, at least in part, occurs through data and related work; but, in contrast to other parts of research data workflows (e.g. collection, processing), the activities of integration are not considered in advance or formalized by scientists. This results in the adoption of informal integration practices that potentially limit the utility of cross-domain collaboration.

“Grunt Work”: Data as Domain

We have seen the ways that data functions (or is supposed to function) as the medium of integration across multiple scholarly domains, but at times, our participants also approached data work as a domain in itself. In these cases, the practice in question became not the integration of data artefacts from multiple domains, but the integration of data work into what is conceived as a non-data-focused, cross-domain scientific workflow. We can make two key observations about this framing of cross-domain data practice.

The first is that scientists differentiate between scientific components of their research and the data work completed in support of that science. We can observe this dichotomy of data work versus ‘real’ research as (for example), Participant C made a distinction between “this group here who can run those pipelines, and these guys up here, the scientists who really are interested in what the results are”, and Participant B commented that among interdisciplinary science labs more generally, “no one remembers that they’re going to need to analyze it until the data comes in, I get so many last minute requests for people to- you know, I- ‘we need this analyzed and out the door and we don’t have anyone for it, do you have anyone to spare?’” A similar difference is made in participants’ workflow sketches, where data-centric steps are labeled with names of software, file types, or words such as ‘pre-processing’ and ‘cleaning’, while later interpretation steps—still reliant on the same data—are more often labeled with names of individual scientists in the lab. In doing so, data work is conceptualized as something automatic and instrumentalized, whereas data *interpretation* as positioned as science completed by thinking, communicating human experts, even when participants made clear verbally that humans are involved at every stage. **Data work is thus framed as incidental to cross-domain science, rather than as part of that science.**

[insert Figure 2]

Fig. 2. A data workflow sketch from Participant A. The red line down the middle of the image divides the lab’s workflow into data processing and modeling as separate parts of the work. Redacted components of image (covered by black rectangles) contain first names of lab members.

The second emerging trend from participants’ conceptions of data-work-as-domain subservient to ‘primary’ research domains was a further hierarchical differentiation between *kinds* of data work. We can see this idea in Figure 2, where Participant A drew a red line dividing their laboratory workflow into data processing and storage (written label as ‘software’) versus model development and analysis. Similarly, Participant B described how, “everybody wants to do the fancy tool development here [pointing at drawing, later stage of research] and nobody wants to run this bit [pointing at drawing, earlier stages of research]... nobody wants to do the grunt work in the middle.” The implicit value judgment made about data ‘grunt work’ even in data-intensive research laboratories is reflective of a widespread devaluation of this work, only being addressed in recent years, that has potentially severe consequences for data quality

and procedural rigour (Jo and Gebru, 2020; D'Ignazio and Klein, 2020; Sambasivan et al., 2021; Muller and Strohmayr, 2022). It is worth noting that Participant C offered a contrasting perspective, stating that the advent of big data and high performance computing resources has fundamentally shifted their primary domain into, “essentially an information science”—this stance on data work was the exception, not the rule.

Discussion

Conclusions

In light of these findings, we can revisit our research questions. With respect to question A, *‘How do scientists’ data practices shape how they integrate across domains?’*, we have observed that data are collected and processed primarily in single-domain workflows, even in interdisciplinary lab settings, and that these siloed data practices result in struggles with integration of the data, which seems to occur only toward the end of research workflows. We have seen that cross-domain integration *is* conceived of by scientists as a data practice, and occurs as one in laboratory settings, but that the activities comprising this practice are not formalized as such. To research question B, then, *‘How do cross-domain research goals shape scientists’ data practices?’*, our answer seems to be that, bluntly, they don’t. Scientists know their labs’ data workflows and practices extremely well, *except* as pertains to cross-domain integration—as evidenced elsewhere in the literature and raised earlier in this article, data work tends to happen according to single-domain standards, considered minimally relevant to the scientific goals of the project, and cross-domain integration is, apparently, a later event that just ‘happens’.

The implications of this failure to articulate and document the activities of cross-domain integration to the same standard that we expect researchers to articulate other research data practices are self-evident: Thomer et al. (2022) make the case that when data work becomes ‘invisible’, it becomes impossible to understand the impact of specific actions on research outputs or on other steps in the research process. This can have adverse impacts on reproducibility (Gundersen, 2021), which is already a particular issue in the context of interdisciplinary and cross-domain research (Wagner et al., 2011) and on the ability of peers and regulatory bodies to evaluate cross-domain research—for example, by subjecting evaluations to outcome bias, where positive outcomes can obfuscate the impact of randomness

or invalid methodological choices (Fanelli, 2012; Brownback and Kuhn, 2019). Moreover, government agencies cite the integration of broad data as a priority (Zogheib and Mahetaji, 2024), and much cross-domain data work is practical in nature and applied to address particular problems such as climate change, social determinants of health, and national security. The minimal attention devoted to integration as a data practice, and the general lack of transparency and accountability for the actions comprising this practice, when the *results* of integration have direct effects on people's lives, health, and privacy, should be a concern to any researcher conscious of the social impacts of their work.

I have demonstrated here that broad data and the integrative work we do with it, like data work more generally, is not incidental. It is part of scientific workflows and impacts the conclusions that scientific research reaches. Thus, in the same way that issues of reproducibility, transparency, and accountability of data science in general are addressed through documentation-based solutions such as model cards (Mitchell et al., 2019) and datasheets (Gebru et al., 2021), I suggest that the most actionable way forward is to prioritize giving cross-domain scientists the tools and vocabulary with which to articulate what they are doing when they say they are integrating across domains. Rather than prescribing how scientists *should* integrate broad data, we can aim (in future work building off this study) to produce scaffolding so that cross-domain and interdisciplinary scientists can document their integrative decisions and activities, improving accountability and transparency of the resulting research outputs. Such a solution could be flexible enough to encompass the wide range of research approaches that fall under the 'cross domain' umbrella, and could also make room for the possibility of integration not as a single 'point' in a data workflow, but as an ongoing, iterative practice, both social and technical aspects of which are understood as essential to how scientists make use of big and broad data.

Limitations

The limited number of sites from a limited geographic region explored here is well within norms for comparable ethnographic research, given the volume of data (drawings, transcripts, and field notes) generated from each interview, but does mean that results are not statistically generalizable to a hypothetical total population of interdisciplinary scientists. This study also focused on scientific laboratories with multiple scientists doing cross-domain work, not on individual scientists doing cross-domain work at a smaller scale or in non-scientific contexts—future work could fill this gap by exploring the cross-domain data practices of humanities-

focused scientists, and by considering integration as a practice undertaken by individual scientists, not necessarily synonymous with collaboration in a laboratory setting.

Finally, all participants in this study are investigators at laboratories affiliated with established research institutions. My focus on data work and cross-domain science in these settings is by no means a suggestion that data work and science are not done outside of institutional contexts: Cross-domain scholarship has strong connections to activist research and research intentionally situated outside of traditional academic settings (Carp et al., 2001; Schmidt, 2022), and this work has incorporated rigorous data practice since its origins (D'Ignazio and Klein, 2020).

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